

Homework #1

Source : [Discretiation and perturbations in simulation of localized turbulence in a pipe with a sudden expansion.](#)

Course : Computational Fluid Dyanamics - IISC

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Q1. What are the workers trying to model ?

They are trying simulate/model, Localized turbulence flow instabilities with the formation of turbulence puffs through a pipe with sudden expansion.

Q2. Is the model in 2d or 3d ?

Its a three dimiensional model, 3-d representation of geometry with the centerline along with variables.

Q3. Did the workers test to see if the solution was grid independent ?

Yes, the test included grids G1,G2,G3,G4,G5,G6 where G1 was finest and G6 least finest.

Q4. Was the solution grid independent?

No. One of the main sources of perturbation that triggers and sustains turbulence is the spatial discretization errorrs in the solution of the equations resulting in simulations with clear grid dependence when comparing G6 and G5 to the others.

Q5. How many control volumes were used in the grid ?

Grids 1 through 6 had different control volumes.

Grid	control Volumes
G1	14628432
G2	8583124
G3	3554560
G4	1506384
G5	828540
G6	643104

Q6. What fluid are they considering ?

The simulations were run assuming incompressible newtonian fluid with poiseuille flow with reynolds numbers varying from 100 to 2500.

Q7. What are the properties of the fluid ?

Constant density and viscosity (incompressible fluid)

Q8. What turbulence models were used ?

Turbulence was not modelled. DNS (Direct numerical simulation) was used for high-fidelity.

Q9. which commercial or open source CFD package have they used ?

The following packages were used among others. [Package|Used for] [-|-] [OpenFOAM v7|CFD package] [Ansys Meshing v 194|Meshing] [OpenMPI|Parallelization of computation]

Q10. What is the method of discretization used ?

The fluid domain was discretized into a computational grid composed of hexahedral cells using cylindrical co-ordinate system transformed from cartesian co-ordinate system. The distribution of grid cells is in radial r and streamwise x directions maintaining aspect ratios across all grids below 10.

Q11. Was convergence achieved ?

The simulation converged with a steady state with criteria of 1×10^{-6} for pressure and velocity

Q12. Where are flow fields visualized ?

In fig 5,6,7 flow fields are visualized for localized turbulence, reattachment distance and recirculation region.

Q13. Was there any validation that was done for the results ?

Yes, existing literature was compared to the results obtained by simulation of Reattachment lengths and perturbation amplitudes.

- Back & Roschke (1972)
- Latonell & Pollard (1986)
- Pak et al. (1990)
- Moallemi & Brinkerhoff (2018)

What was the measurement technique used ? DNS

What was the final comment on the agreement ?

Turbulence that seems to arise naturally in numerical simulations has probably been triggered by the amplification of disturbances caused by numerical error. Perfectly unperturbed simulations without any numerical errors or disturbances would result in fully laminar solutions, regardless of Reynolds number. The main sources of numerical disturbances identified in the present study were from spatial and temporal discretization errors.